

1 Perception

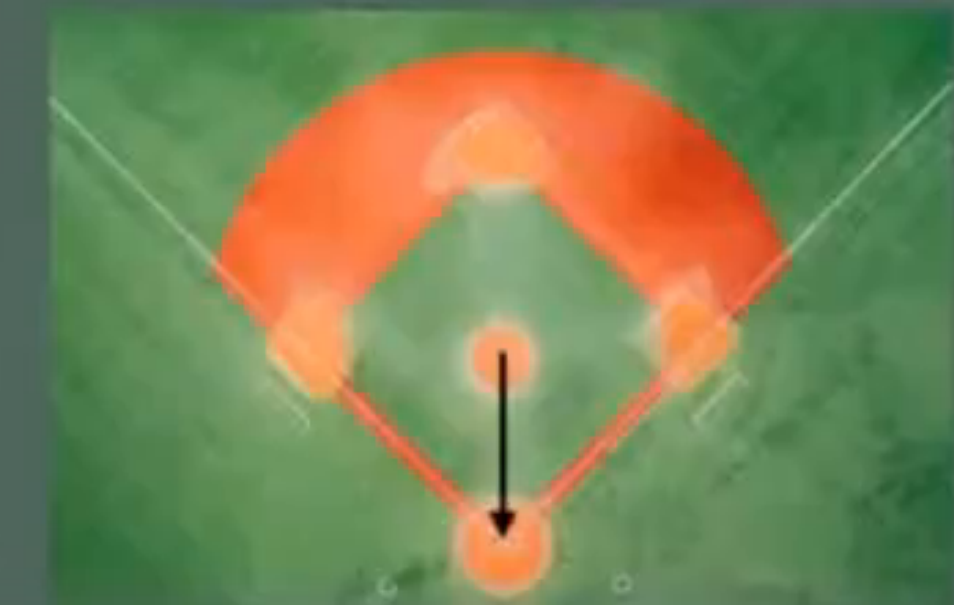
The limits of human perception.

- The "four-seam" Fastball.
 - How to Throw It
 - How to Hit It
- The Limits of Human Perception
 - Reality
 - Illusion
 - Cognitive Dissonance
- Lessons Learned
 - Sometimes Seeing is NOT Believing
 - Prediction, Adaptation, Intuition, and Critical Thinking



Things to know

- The distance from the pitcher's mound to home plate is 60 feet 6 inches.
- The fastest recorded four-seam fastball was clocked at 108 miles per hour and thrown by Nolan Ryan.
- The pitcher must release the ball within a one-millisecond boundary to hit the strike zone. (< 1ms or > 1ms results in a miss.)
- Five milliseconds are required to instantiate release of the ball. **The pitcher is predicting the release point with millisecond accuracy!**
- A 108-mph fastball travels at 158.4 ft. per second. (Does not account for actual release point.)
- The time required for a 108-mph fastball to reach home plate, assuming a 5 ft reduction for release point, is 0.395 seconds.
- The batter must see the ball, determine its trajectory, initiate the swing, and complete the follow-through within 0.395 seconds.
- This is beyond the perceptual capabilities of most humans, including professional baseball players!
- The batter faces two significant hurdles in perception.
 - The reverse spin of the threads on the ball causes the ball to fall slower than the batter predicts. This actual trajectory is higher than expected.
 - The human visual cortex is unable to process an object moving at this speed over such a short distance, and the ball appears to rise.
- **The batter is swinging based on an educated guess and must actively ignore perception! They can't see what they are trying to hit**



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